CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

Androgen receptor axis: 7 molecules stopped early, all small molecules

13 stopped trials, but 7 molecules — all small molecules: Enzalutamide, Abiraterone, Abiraterone Acetate, Apalutamide and 3 more. 10 of the stops were the trial's own verdict and 1 followed a decision taken elsewhere; 2 cannot be established. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

10.6%
13 of 123 closed trials (95% CI 6.3%–17.2%)

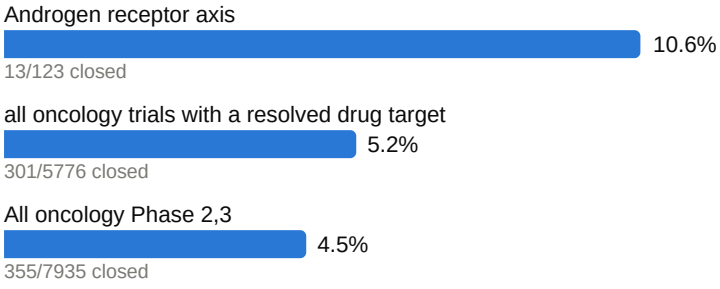
10 / 3 / 0
efficacy / safety / both (adds to 13)

41.8%
of 294 trials have closed · 171 still open or unresolved

4.5%
stopped for a biological reason within 3 years of starting (95% CI 2.4%–8.2%; 161 still at risk; 1.5% area-wide)

1 of 5 **Completed and missed the primary endpoint.** Of 78 completed trials in this segment, 5 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 1 missed and 4 met. Counted separately and never in the rate above; a floor, not a rate. 29 of the 76 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 190 distinct experimental drugs across 294 trials (small molecule 287, unknown 47, protein 41, antibody 35). Most tested: ENZALUTAMIDE (99), APALUTAMIDE (57), ABIRATERONE ACETATE (53), PREDNISONE (52), DAROLUTAMIDE (43). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 13 stops came from 13 sponsor-asset programmes across 13 sponsors; the largest (SOLTI Breast Cancer Research Group) contributed 1; removing that programme's trials from both sides leaves 9.8%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 6 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 15.4%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	3	60	5.0%	1.7%–13.7%
2018–2020	8	41	19.5%	10.2%–34.0%
2021–2024	2	22	9.1%	2.5%–27.8%

Where the stops sit

Efficacy / futility: 10, Safety: 3.
Sponsors with most stops: Mayo Clinic (1), Epizyme, Inc. (1), ESSA Pharmaceuticals (1), Institute of Cancer Research, United Kingdom (1), SOLTI Breast Cancer Research Group (1), Bayer (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05689021	2	Mayo Clinic	ABIRATERONE ACETATE, NIRAPARIB, PREDNISONE	2026-07	Efficacy	Sponsor decided to close due to slow accrual and lack of efficacy
NCT04179864	2	Epizyme, Inc.	TAZEMETOSTAT, ABIRATERONE, PREDNISONE	2024-11	Efficacy	The study was terminated early as Sponsor decided to discontinue the development of tazemetostat in metastatic castration resistant prostate cancer and the primary endpoint was not met for this study. There were no safety concerns.
NCT05075577	2	ESSA Pharmaceuticals	MASOFANITEN, ENZALUTAMIDE	2024-10	Efficacy	Study was terminated early due to a lack of improved efficacy.
NCT04458311	2	Institute of Cancer Research, United Kingdom	ABIRATERONE ACETATE, TILDRAKIZUMAB	2023-04	Efficacy	Funder decided that response outcomes to date did not support continuation of the study.
NCT04142060	2	SOLTI Breast Cancer Research Group	ENZALUTAMIDE	2022-10	Efficacy	Lack of effectiveness
NCT04157088	2	Bayer	DAROLUTAMIDE, ENZALUTAMIDE	2022-07	Efficacy	The results of the lead-in phase did not fulfill the criteria set for moving into the randomized phase

The 6 most recent of 13 stopped trials. Every molecule behind all 13 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 41.8% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.